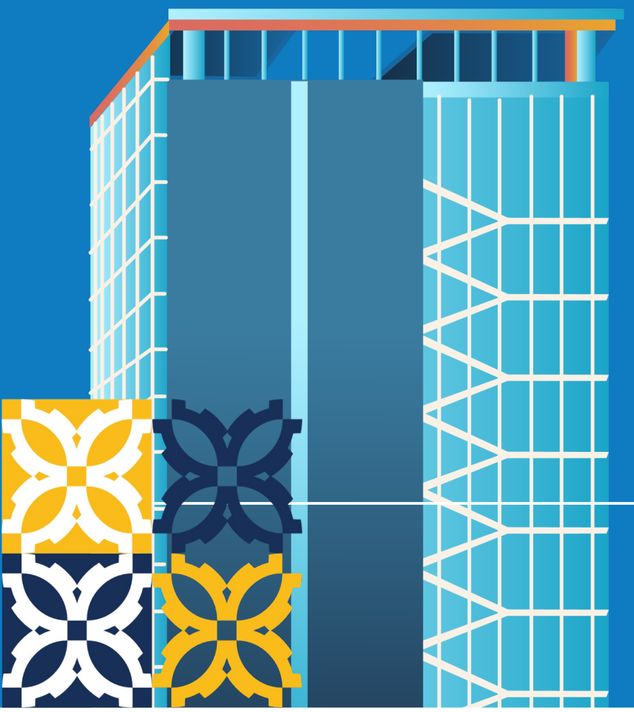


Direct Manipulation

Lecture 6

Tim RMK GIGA
Departemen Teknik Informatika

Direct Manipulation

- This principle suggests that users should feel as though they are directly controlling the objects they are working with.
- For example, dragging an image to resize it on a touchscreen is more intuitive than pressing a button to zoom in.
- This approach enhances user engagement by allowing for more natural interactions.

Invisible Interfaces

- The best interfaces are those that become invisible to the user, allowing them to focus entirely on the task at hand rather than the interface itself.
- An invisible interface minimizes the cognitive load, enabling users to concentrate on their objectives without being distracted by the technology.

It is important to design interfaces that facilitate seamless interaction, making technology feel more intuitive and user-friendly.

The Desktop Metaphor 1

The desktop metaphor mimics physical files and folders, allowing users to interact with digital objects in a way that feels natural and intuitive.

Gulf of Execution and Evaluation: The goal is to narrow the gap between what users want to do (execution) and what they perceive they can do (evaluation).

Direct Manipulation: This interaction style allows users to directly interact with objects on the screen, such as dragging files to move them, which mirrors real-world actions.

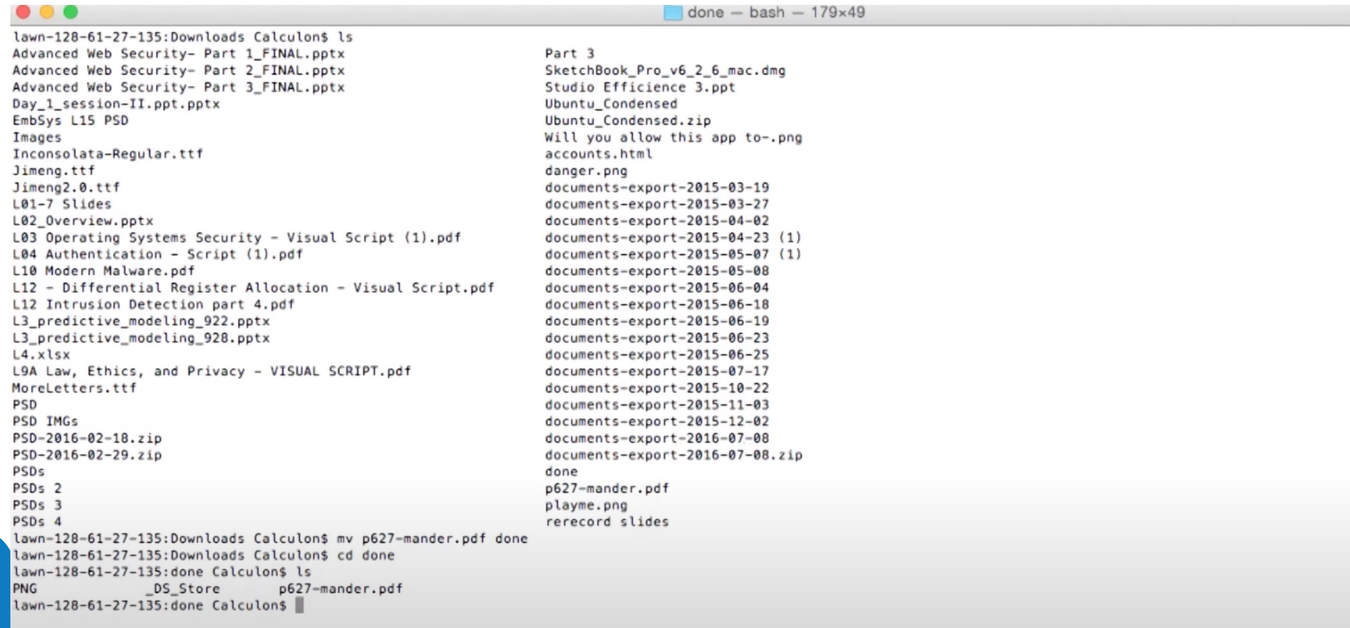
Intuitive Interaction: By leveraging real-world experiences, direct manipulation makes it easier for users to understand and perform tasks on a computer, enhancing usability.

It is important to design interfaces that facilitate natural interaction, making technology feel more accessible and user-friendly.

The Desktop Metaphor 2

The desktop metaphor mimics physical files and folders, allowing users to interact with digital objects in a way that feels natural and intuitive.

Historical Context: Before graphical user interfaces, users relied on command line interfaces, which required memorizing commands and offered limited feedback.



The Desktop Metaphor 2

Evolution of Interaction: The introduction of the mouse allowed for more direct manipulation, enabling users to click and drag files, which felt more natural and provided instant feedback.

Touchscreens: Modern touchscreens have further enhanced direct manipulation, allowing users to interact with icons and files in a way that closely resembles physical actions.

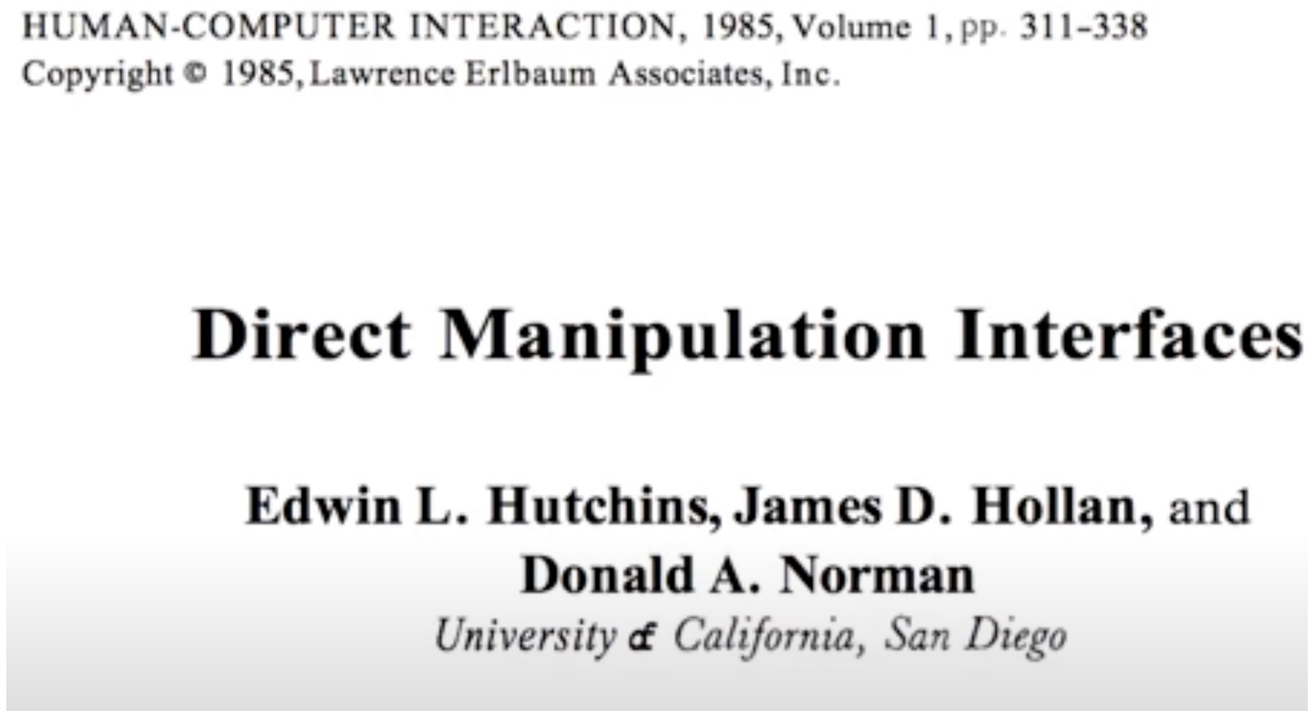
It is important to design interfaces that allow users to interact as directly as possible with their tasks, minimizing the gap between intention and action.

Paper "Direct Manipulation Interfaces"

A seminal paper from 1985 co-authored by Edwin Hutchins, James Hollan, and Don Norman, introduced key ideas about how users interact with systems.

There are two aspects of directness:

distance and **direct engagement**


HUMAN-COMPUTER INTERACTION, 1985, Volume 1, pp. 311-338
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Direct Manipulation Interfaces

Edwin L. Hutchins, James D. Hollan, and
Donald A. Norman
University of California, San Diego

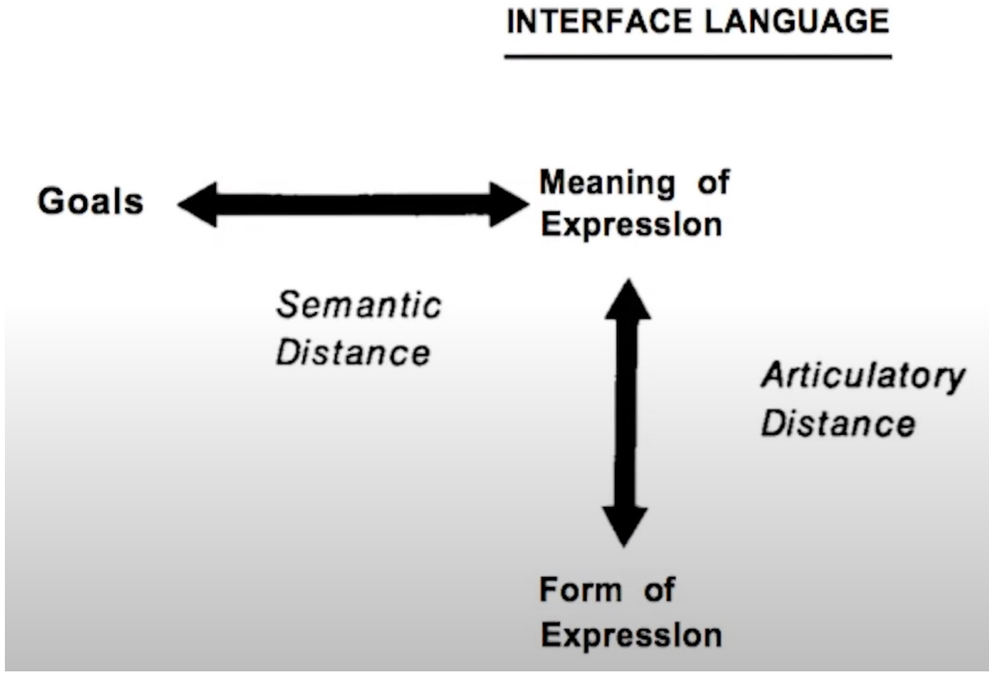
Paper "Direct Manipulation Interfaces"

Distance: the distance between user's goals and the system itself.

Two types of distance:

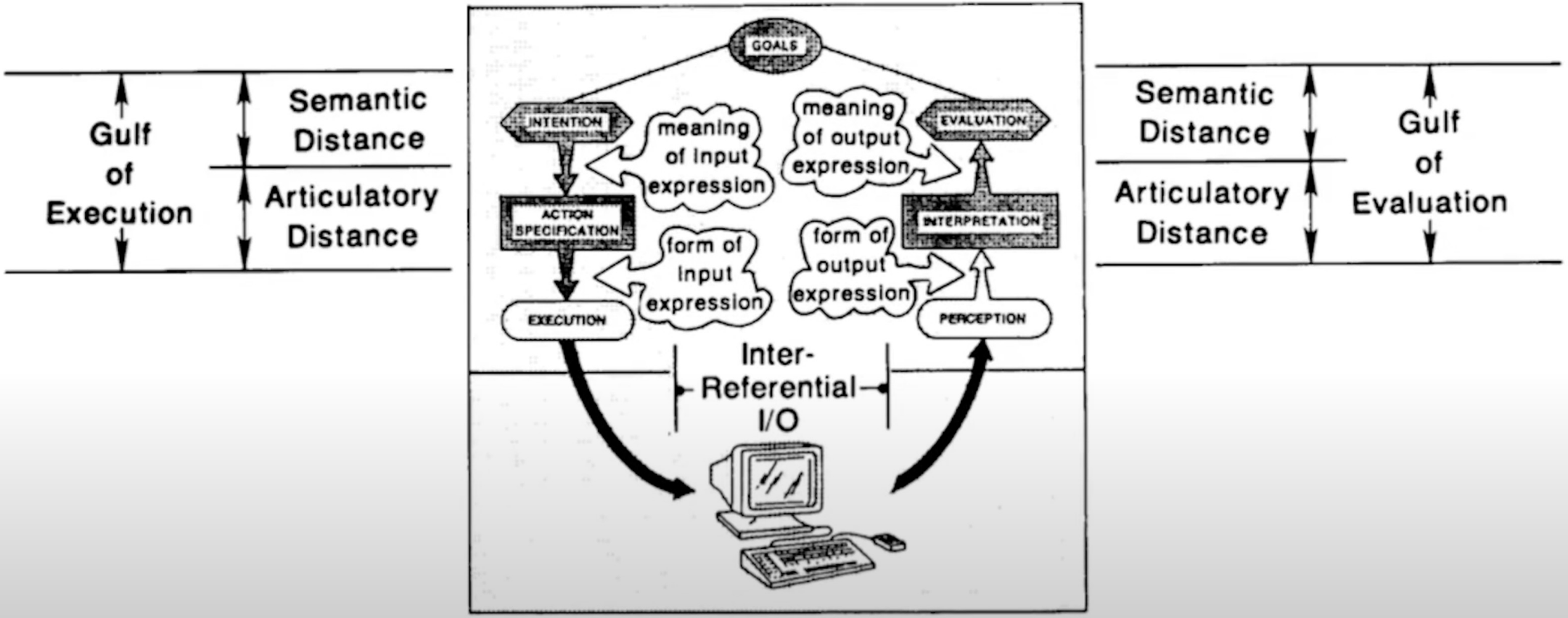
semantic distance (the difference between user goals and their expression in the system) --> how hard it is to know what to do

articulatory distance (the gap between expression and execution). --> how hard it is to actually do what you know to do



"The feeling of directness is inversely proportional to the cognitive effort required to manipulate and evaluate the system." --> Greater effort means less directness

Paper "Direct Manipulation Interfaces"



Paper "Direct Manipulation Interfaces"

Feedback Cycles:

The interaction process involves

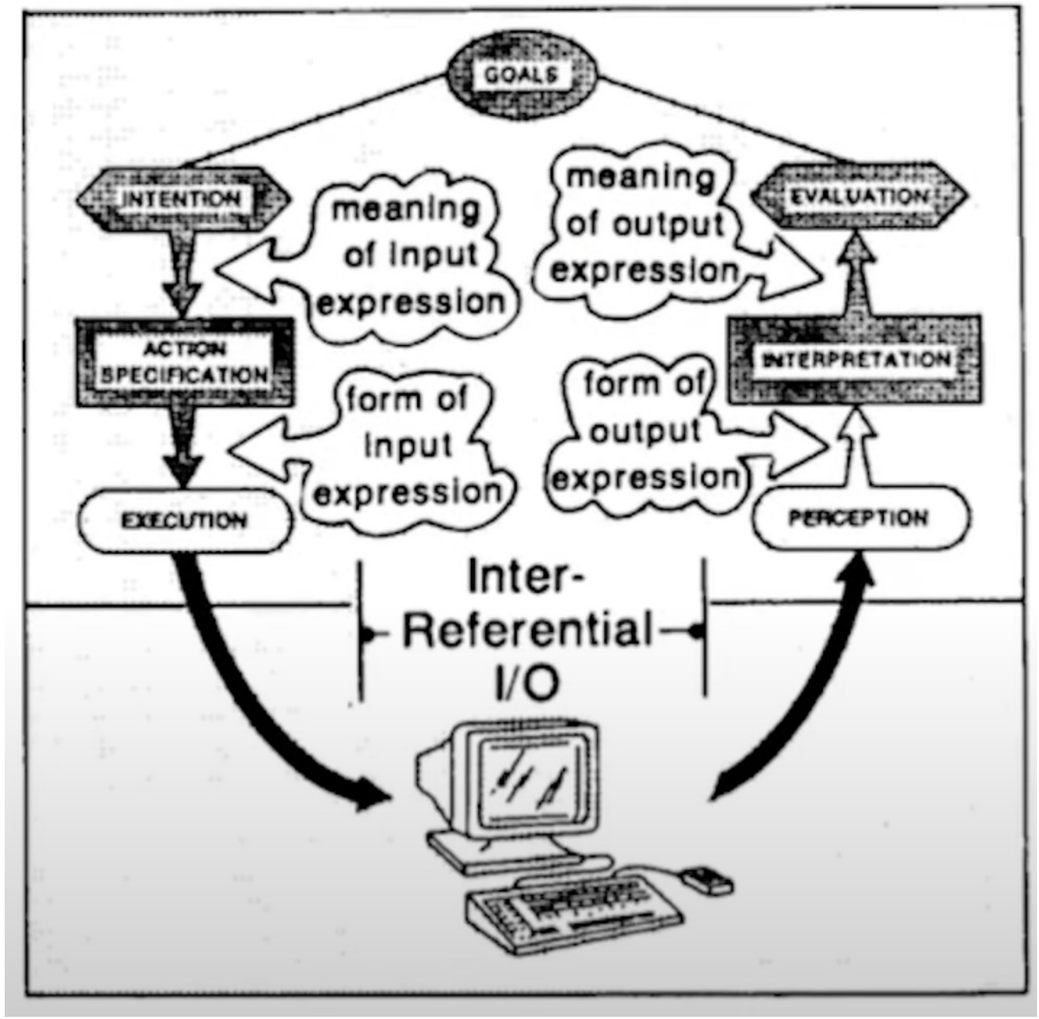
users setting goals,

translating them into intentions,

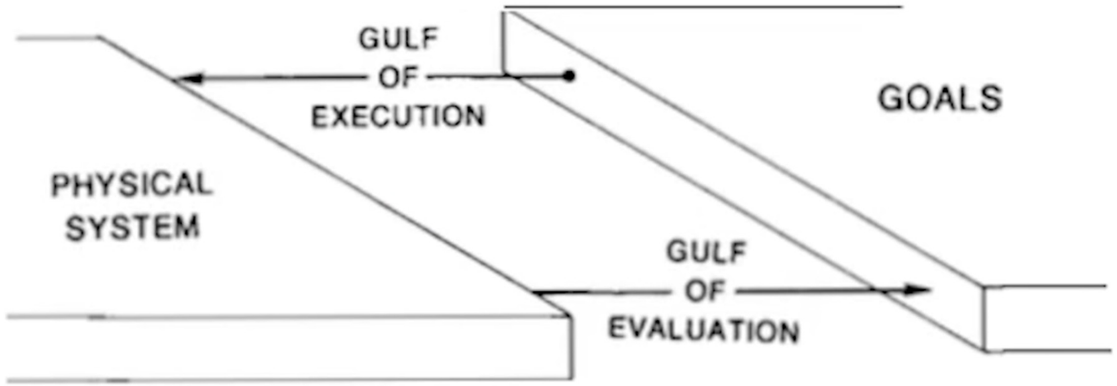
executing actions, and

evaluating the system's output.

This cycle is crucial for effective interaction.



Paper "Direct Manipulation Interfaces"



Direct Engagement

Direct manipulation allows users to **feel directly engaged with the objects of their goals** rather than the computer itself.

This is illustrated through examples like moving files or controlling characters in a game.

Direct manipulation is a powerful method for shortening "the distance".

It is important to design interfaces by minimizing cognitive load and enhancing user engagement through direct manipulation interfaces.

Exploring HCI: Direct Manipulation & VR

The advancements in virtual reality (VR) facilitate direct manipulation in design tasks that were previously challenging, such as in 3D Design.

Traditionally, designers used 2D interfaces to create 3D objects, which complicated the direct manipulation of designs.



Exploring HCI: Direct Manipulation & VR

Benefits of VR

VR allows designers to view and interact with their creations in a 3D space, enabling them to manipulate objects as if they were physically present.

This includes rotating and moving objects using natural hand motions.

Feedback Issues

The challenges of providing tactile feedback in VR, as users need to feel a sense of physical interaction when manipulating virtual objects

There is a potential of VR to enhance the design process through direct manipulation, while acknowledging that there are still challenges to overcome in creating a fully immersive experience.

Babies and Direct Manipulation

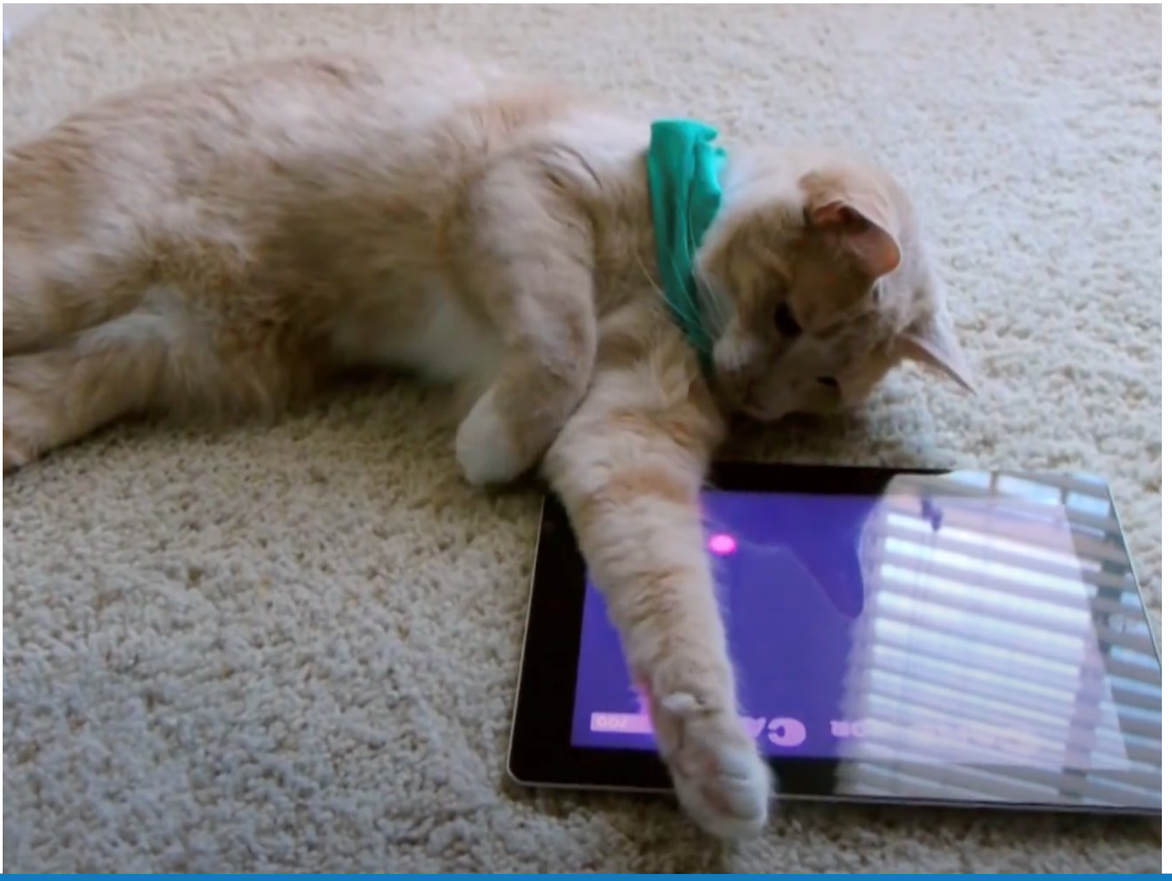


The power of direct manipulation interfaces through the example of a young child using a tablet.

A baby named Lucy, who, at 18 months old, is able to interact with a Kindle Fire tablet effectively.

Lucy can use the tablet because it allows for direct interaction with the screen, unlike traditional input methods such as a keyboard or mouse, which she is not yet capable of using.

Babies and Direct Manipulation



Direct manipulation is not limited to humans; it even extends to animals, such as cats, which can engage with tablet games designed for them.

This illustrates the intuitive nature of direct manipulation interfaces, making technology accessible to users of all ages and backgrounds.

Making Indirect Manipulation Direct

Goal: how to enhance user interactions with technology by making indirect manipulations feel more direct.

Example: the Mac Touchpad, where gestures like swiping with two fingers to open the notification center create a sense of directness because the action corresponds closely to the location on the screen.

Making Indirect Manipulation Direct

Spatial Correspondence: when actions on an input device (like a touchpad) correspond directly to visual changes on the screen, users feel a stronger sense of control.

Animation's Role: animations can reinforce the feeling of direct manipulation. For instance, spreading fingers on the touchpad to clear windows is enhanced by animations that mimic the action, making it feel more intuitive.

Design Choices: designers have the flexibility to create interactions that feel direct, and the choice of animations and spatial arrangements can significantly impact user experience.

It is important to design interactions that minimize the cognitive gap between user actions and system responses, enhancing the overall user experience.

Exploring HCI: Direct Manipulation

Direct manipulation: the way users interact with digital objects in a way that feels immediate and intuitive, similar to how they would interact with physical objects.

Immediate Feedback: Direct manipulation relies on providing users with immediate feedback that corresponds directly to their actions, enhancing their sense of control and engagement.

Gesture-Based Interaction: There are challenges of implementing direct manipulation in gesture-based interfaces, such as virtual reality, where users may not physically touch objects but still need to feel a sense of direct interaction.

Applications: how the principles of direct manipulation can be applied to their chosen area of HCI, considering how to create more intuitive and engaging user experiences.

It is important to design interfaces that facilitate direct manipulation to improve user interaction and satisfaction.

Invisible Interfaces

Invisible interface: the concept of making the interface between users and tasks as seamless as possible, so that users feel as though they are interacting directly with the task rather than through an interface.

Examples: Compare the experience of using a stylus on a tablet, which feels natural and direct, with the less intuitive experience of using a mouse for drawing. This illustrates how different interfaces can affect user perception and interaction.

It is important to design interfaces that minimize cognitive load and enhance user engagement, making technology feel more accessible and user-friendly.

Good Design, Bad Design Invisible Interfaces

Good design: creates interfaces that feel "invisible" to users, allowing them to focus on their tasks

Bad design: interfaces are overly complicated or distracting

Invisible Interfaces: Good design allows users to interact with technology seamlessly, making the interface feel like a natural extension of their actions.

Feedback Importance: Providing clear feedback is essential in ensuring users understand the results of their interactions, which helps maintain engagement and satisfaction.

Invisibility by Learning

An interface can become "invisible" not only through great design but also through user familiarity and learning.

User Learning: As users gain experience and practice with an interface, they become more comfortable and can use it instinctively, making the interface feel less present.

Example of Driving: A driver instinctively reacts to situations without consciously thinking about the controls, demonstrating how practice can lead to a seamless interaction.

Design Expectations: While some tasks, like driving, may warrant extensive learning, most interfaces should be designed to require minimal learning time from users, ideally making them feel integrated into the task.

It is important to design interfaces that facilitate learning and become intuitive over time, enhancing user experience without overwhelming them with complexity.

Invisibility by Design

Goal: creating user interfaces that are "invisible" to users, meaning that users should feel as though they are directly interacting with the task at hand rather than the interface itself.

User Experience: When interfaces are designed well, users are not aware of the interface; they focus solely on their tasks.

Complexity of HCI: How achieving invisibility in design is a challenging goal.

Design Principles: Principles for creating invisible interfaces include leveraging users' prior expectations and providing quick feedback to enhance usability.

5 Tips: Invisible Interfaces

Use Affordances: Design elements should visually indicate how they are meant to be used (e.g., buttons that look clickable).

Know Your User: Understand that different users (novices vs. experts) have varying expectations and needs regarding visibility and interaction.

Differentiate Your User: Provide multiple ways to accomplish tasks to cater to both novice and expert users, ensuring that options are discoverable while also allowing for efficient shortcuts.

Let Your Interface Teach: Design the interface to guide users in learning how to use it effectively, such as displaying keyboard shortcuts alongside menu options.

Talk to Your User: Engage with users to gather feedback on their experiences and thoughts while using the interface, which can help identify areas for improvement.

Summary

Direct Manipulation: Modern technologies, such as touchscreens and virtual reality, enhance the feeling of direct manipulation for users.

Invisible Interfaces: The most effective interfaces become "invisible," allowing users to focus entirely on their tasks without being distracted by the interface itself.

Usable Design: The goal of usable design is to create interfaces that seamlessly integrate with user tasks, minimizing cognitive load and enhancing overall user experience.

Thank You

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